Experiment 4: Ensemble Prediction and Decision Tree Model Evaluation

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SEC : CSE-A

In [ ]:

```

```

1) Aim and Objectives

Aim

- To design, train, and evaluate multiple machine learning classification models for breast cancer prediction, and to compare their performance using standard evaluation metrics such as Accuracy, Precision, Recall, F1-score, and ROC-AUC.

Objectives

- To load and preprocess the breast cancer dataset, including feature scaling and data splitting.
- To train different classification models such as Decision Tree, AdaBoost, Gradient Boosting, XGBoost, Random Forest, and a Stacked Ensemble model.
- To tune model hyperparameters using cross-validation techniques for improved performance.
- To evaluate and compare all models using performance metrics including Accuracy, Precision, Recall, F1-score, and ROC-AUC.
- To visualize model performance using ROC curves and confusion matrices.
- To analyze and identify the best-performing model based on validation and test results.

2)Libraries Used

In [29]:
```
import pandas as pd
import numpy as np
from sklearn.datasets import load_breast_cancer
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import (
    AdaBoostClassifier,
    GradientBoostingClassifier,
    RandomForestClassifier,
    StackingClassifier
)

from sklearn.svm import SVC
from sklearn.naive_bayes import GaussianNB
from sklearn.model_selection import GridSearchCV, cross_val_score
from sklearn.metrics import (
    accuracy_score,
    precision_score,
    recall_score,
    f1_score,
    roc_auc_score,
    roc_curve,
    confusion_matrix
)
from xgboost import XGBClassifier
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay
import numpy as np
import pandas as pd

from sklearn.model_selection import GridSearchCV, RandomizedSearchCV, StratifiedKFo
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier, AdaBoostClassifier, GradientBo
from sklearn.linear_model import LogisticRegression
from xgboost import XGBClassifier
from sklearn.model_selection import cross_val_predict
from sklearn.metrics import roc_curve, auc

data = load_breast_cancer()


df = pd.DataFrame(data.data, columns=data.feature_names)


df['diagnosis'] = data.target   # 0 = malignant, 1 = benign


df.head()
df.tail()
```

Out[29]:

	mean radius	mean texture	mean perimeter	mean area	mean smoothness	mean compactness	mean concavity	mean concave points	sy
564	21.56	22.39	142.00	1479.0	0.11100	0.11590	0.24390	0.13890	
565	20.13	28.25	131.20	1261.0	0.09780	0.10340	0.14400	0.09791	
566	16.60	28.08	108.30	858.1	0.08455	0.10230	0.09251	0.05302	
567	20.60	29.33	140.10	1265.0	0.11780	0.27700	0.35140	0.15200	
568	7.76	24.54	47.92	181.0	0.05263	0.04362	0.00000	0.00000	

5 rows × 31 columns

In [30]:
```
df.isnull().sum()
```

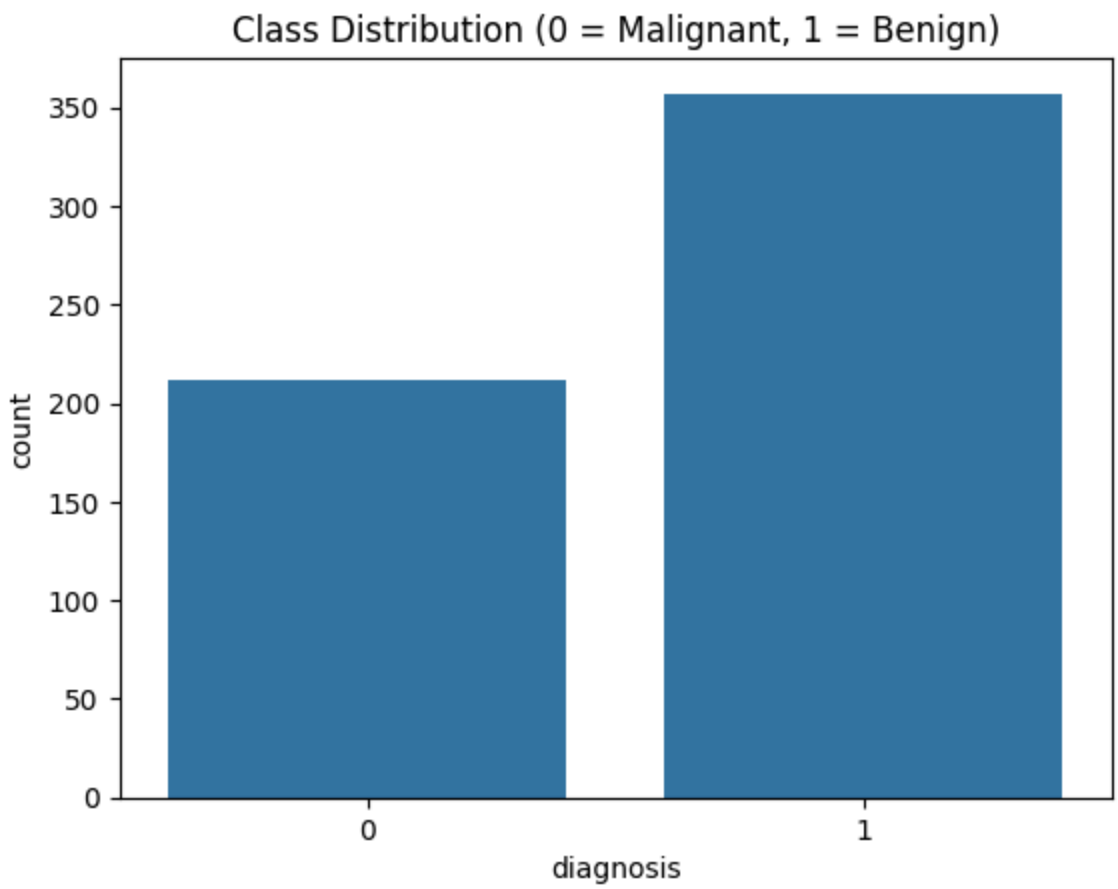
Out[30]:
```
mean radius                0
mean texture               0
mean perimeter             0
mean area                  0
mean smoothness            0
mean compactness           0
mean concavity             0
mean concave points        0
mean symmetry              0
mean fractal dimension     0
radius error               0
texture error              0
perimeter error            0
area error                 0
smoothness error           0
compactness error          0
concavity error            0
concave points error       0
symmetry error             0
fractal dimension error    0
worst radius               0
worst texture              0
worst perimeter            0
worst area                 0
worst smoothness           0
worst compactness          0
worst concavity            0
worst concave points       0
worst symmetry             0
worst fractal dimension    0
diagnosis                  0
dtype: int64
```

In [31]:
```
X = df.drop('diagnosis', axis=1)
y = df['diagnosis']
```

In [32]:
```
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

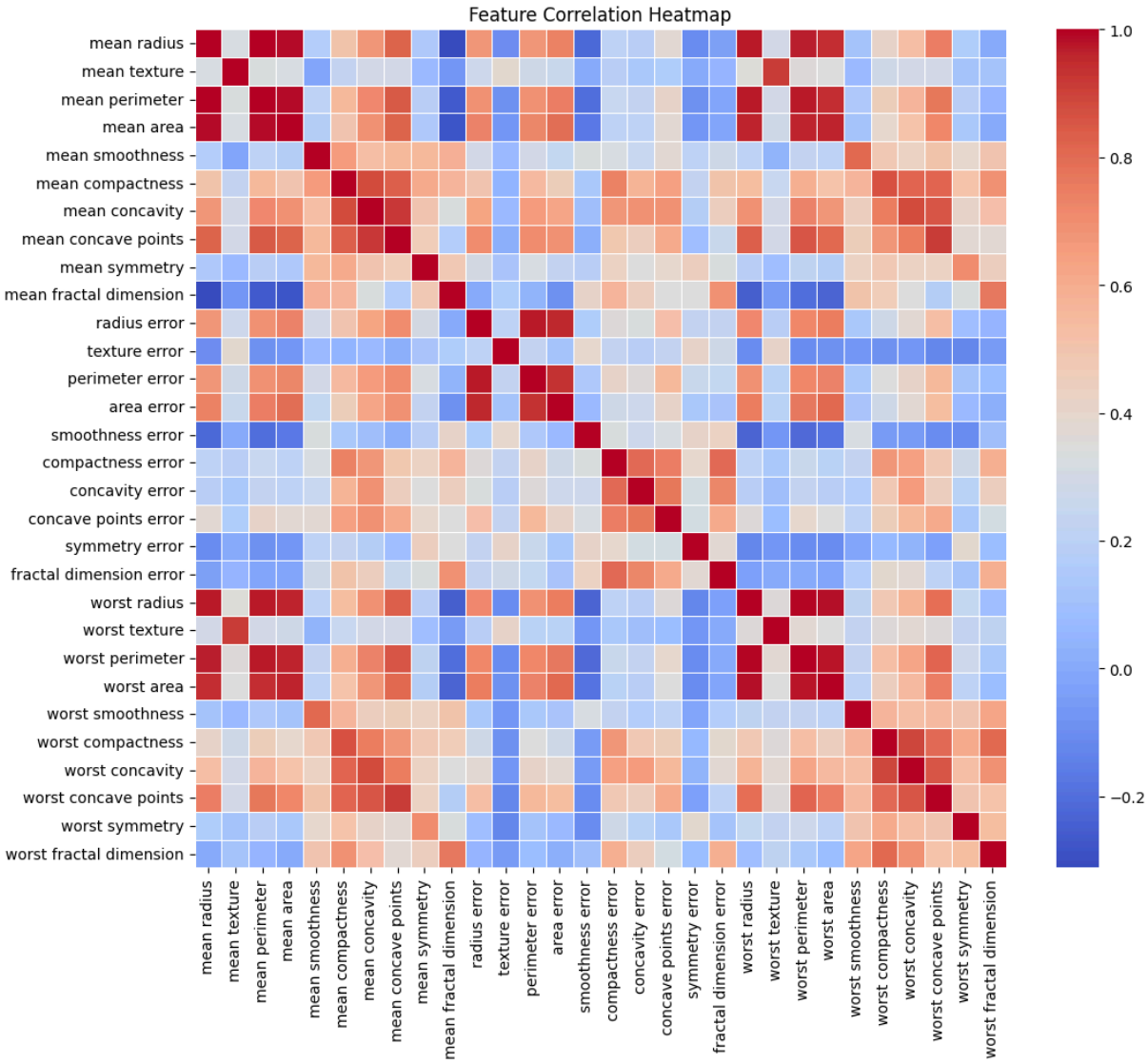
3)Exploratory Data Analysis (EDA)

In [34]:
```
import matplotlib.pyplot as plt
import seaborn as sns

sns.countplot(x=y)
plt.title("Class Distribution (0 = Malignant, 1 = Benign)")
plt.show()
```



In [ ]:

In [35]:
```
plt.figure(figsize=(12,10))
corr = pd.DataFrame(X_scaled, columns=X.columns).corr()

sns.heatmap(corr, cmap='coolwarm', linewidths=0.5)
plt.title("Feature Correlation Heatmap")
plt.show()
```



In [36]:
```
X_scaled = pd.DataFrame(X_scaled, columns=X.columns)

corr_matrix = X_scaled.corr()


corr_matrix.head()

df_corr = X_scaled.copy()
df_corr['diagnosis'] = y


corr_with_target = df_corr.corr()['diagnosis'].sort_values(ascending=False)

corr_with_target
```

Out[36]:
```
diagnosis                  1.000000
smoothness error           0.067016
mean fractal dimension     0.012838
texture error              0.008303
symmetry error             0.006522
fractal dimension error   -0.077972
concavity error           -0.253730
compactness error         -0.292999
worst fractal dimension   -0.323872
mean symmetry             -0.330499
mean smoothness           -0.358560
concave points error      -0.408042
mean texture              -0.415185
worst symmetry            -0.416294
worst smoothness          -0.421465
worst texture             -0.456903
area error                -0.548236
perimeter error           -0.556141
radius error              -0.567134
worst compactness         -0.590998
mean compactness          -0.596534
worst concavity           -0.659610
mean concavity            -0.696360
mean area                 -0.708984
mean radius               -0.730029
worst area                -0.733825
mean perimeter            -0.742636
worst radius              -0.776454
mean concave points       -0.776614
worst perimeter           -0.782914
worst concave points      -0.793566
Name: diagnosis, dtype: float64
```

4) Train–Test Split (80–30)

In [37]:
```python
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X_scaled,
    y,
    test_size=0.3,
    random_state=42,
    stratify=y
)

print("Training samples:", X_train.shape)
print("Testing samples:", X_test.shape)
```

```
Training samples: (398, 30)
Testing samples: (171, 30)
```

5)Training models

Decision Tree

In [38]:
```
from sklearn.tree import DecisionTreeClassifier

dt = DecisionTreeClassifier(
    criterion="gini",
    max_depth=3,
    random_state=42
)

dt.fit(X_train, y_train)
```

Out[38]:
```
▾ DecisionTreeClassifier
▶ Parameters
```

AdaBoost

In [39]:
```
from sklearn.ensemble import AdaBoostClassifier

ada = AdaBoostClassifier(
    n_estimators=200,
    learning_rate=0.1,
    random_state=42
)

ada.fit(X_train, y_train)
```

Out[39]:
```
▾ AdaBoostClassifier
▶ Parameters
```

Gradient Boosting

In [40]:
```
from sklearn.ensemble import GradientBoostingClassifier

gb = GradientBoostingClassifier(
    n_estimators=200,
    learning_rate=0.1,
    max_depth=3,
    random_state=42
)

gb.fit(X_train, y_train)
```

Out[40]:
```
▾ GradientBoostingClassifier
▶ Parameters
```

XGBoost

In [41]:
```
from xgboost import XGBClassifier

xgb = XGBClassifier(
    n_estimators=200,
    learning_rate=0.1,
    max_depth=5,
    subsample=0.8,
    eval_metric="logloss",
    random_state=42
)

xgb.fit(X_train, y_train)
```

Out[41]:
```
▾ XGBClassifier
▶ Parameters
```

Random Forest

In [42]:
```
from sklearn.ensemble import RandomForestClassifier

rf = RandomForestClassifier(
    n_estimators=100,
    max_depth=None,
    criterion="gini",
    random_state=42
)

rf.fit(X_train, y_train)
```

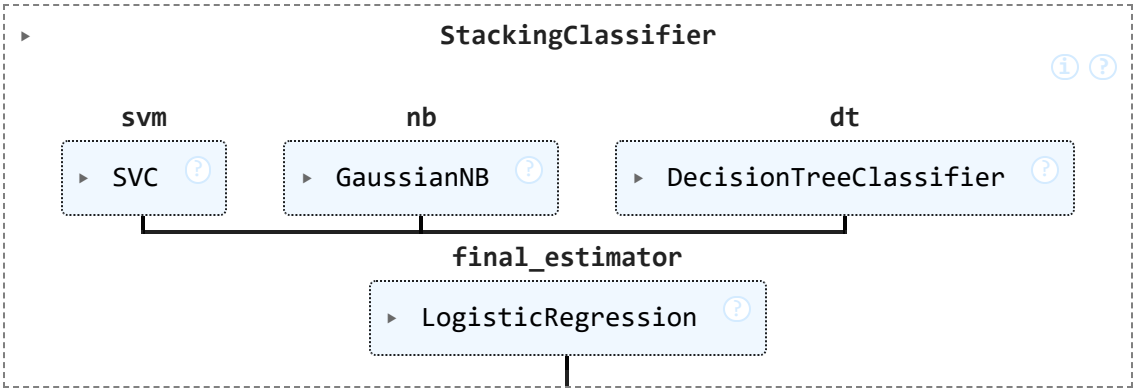
Out[42]:
```
▾ RandomForestClassifier
▶ Parameters
```

Stacking Classifier

In [43]:
```
from sklearn.svm import SVC
from sklearn.naive_bayes import GaussianNB
from sklearn.ensemble import StackingClassifier
from sklearn.linear_model import LogisticRegression

estimators = [
    ("svm", SVC(probability=True, kernel="rbf")),
    ("nb", GaussianNB()),
    ("dt", DecisionTreeClassifier(max_depth=3))
]
```

```
stack = StackingClassifier(
    estimators=estimators,
    final_estimator=LogisticRegression(),
    cv=5
)

stack.fit(X_train, y_train)
```

Out[43]:



6)Confusion Matrix and ROC

Decision Tree

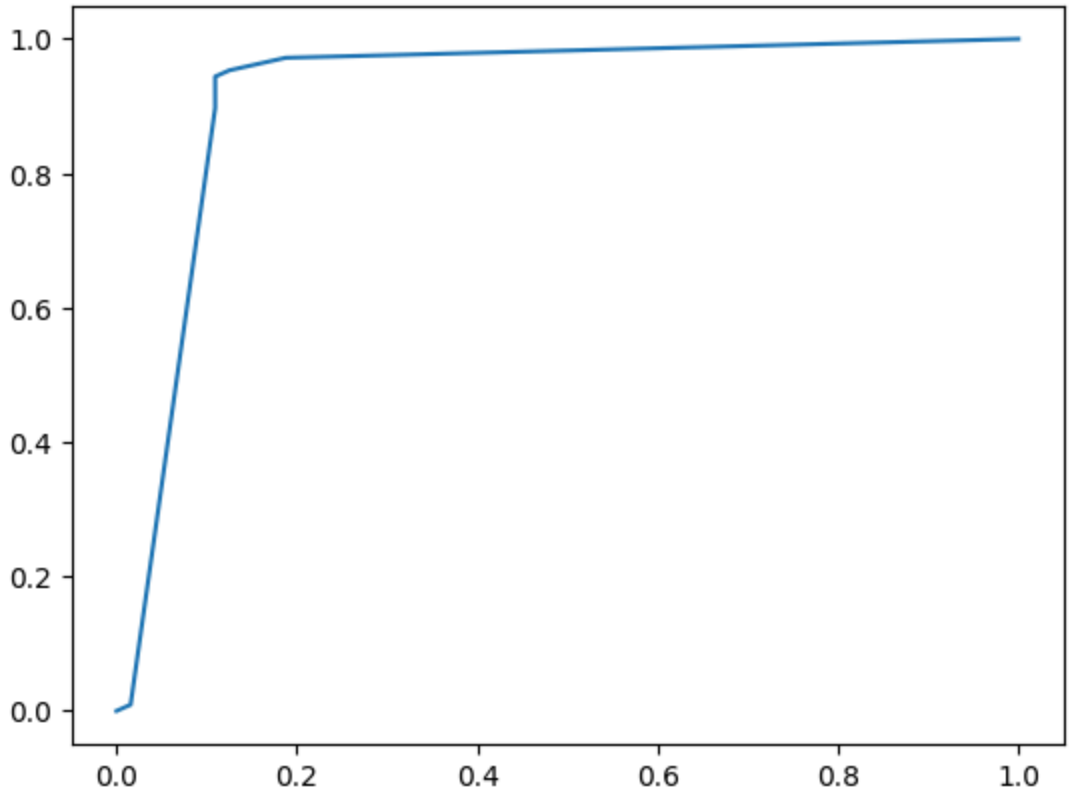
In [44]:

```
def evaluate_model(model, X_test, y_test, name):
    y_pred = model.predict(X_test)
    y_prob = model.predict_proba(X_test)[:, 1]

    print(f"\n{name}")
    print("Accuracy :", accuracy_score(y_test, y_pred))
    print("Precision:", precision_score(y_test, y_pred))
    print("Recall   :", recall_score(y_test, y_pred))
    print("F1 Score :", f1_score(y_test, y_pred))
    print("ROC AUC  :", roc_auc_score(y_test, y_prob))

    fpr, tpr, _ = roc_curve(y_test, y_prob)
    plt.plot(fpr, tpr, label=name)
```

In [45]:

```
dt = DecisionTreeClassifier(random_state=42)

dt_params = {
    'criterion': ['gini', 'entropy'],
    'max_depth': [3, 5, 10, None],
    'min_samples_split': [2, 5, 10]
}

dt_grid = GridSearchCV(dt, dt_params, cv=5, scoring='accuracy')
dt_grid.fit(X_train, y_train)

best_dt = dt_grid.best_estimator_
print("Best Decision Tree Params:", dt_grid.best_params_)

evaluate_model(best_dt, X_test, y_test, "Decision Tree")
```

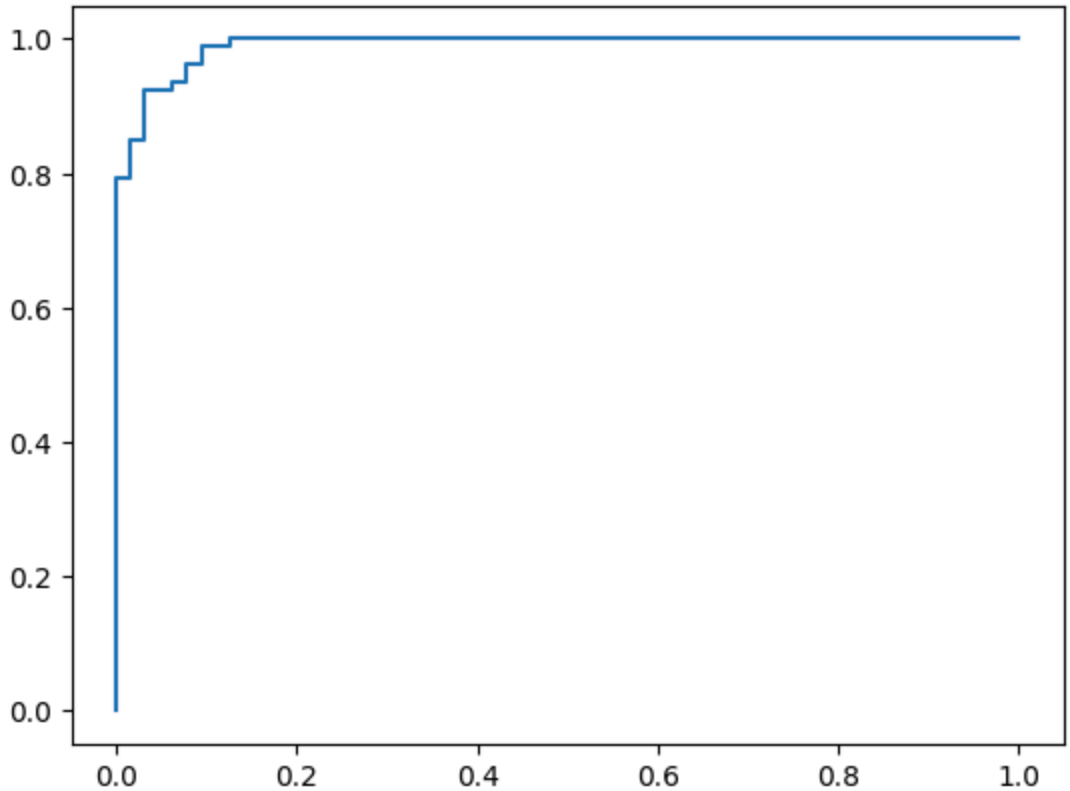
```
Best Decision Tree Params: {'criterion': 'gini', 'max_depth': 3, 'min_samples_spli
t': 2}

Decision Tree
Accuracy : 0.9239766081871345
Precision: 0.9272727272727272
Recall   : 0.9532710280373832
F1 Score : 0.9400921658986175
ROC AUC  : 0.9186623831775702
```



AdaBoost

In [46]:
```
ada = AdaBoostClassifier(random_state=42)

ada_params = {
    'n_estimators': [50, 100, 200],
    'learning_rate': [0.01, 0.1, 1]
}

ada_grid = GridSearchCV(ada, ada_params, cv=5, scoring='accuracy')
ada_grid.fit(X_train, y_train)

best_ada = ada_grid.best_estimator_
print("Best AdaBoost Params:", ada_grid.best_params_)

evaluate_model(best_ada, X_test, y_test, "AdaBoost")
```

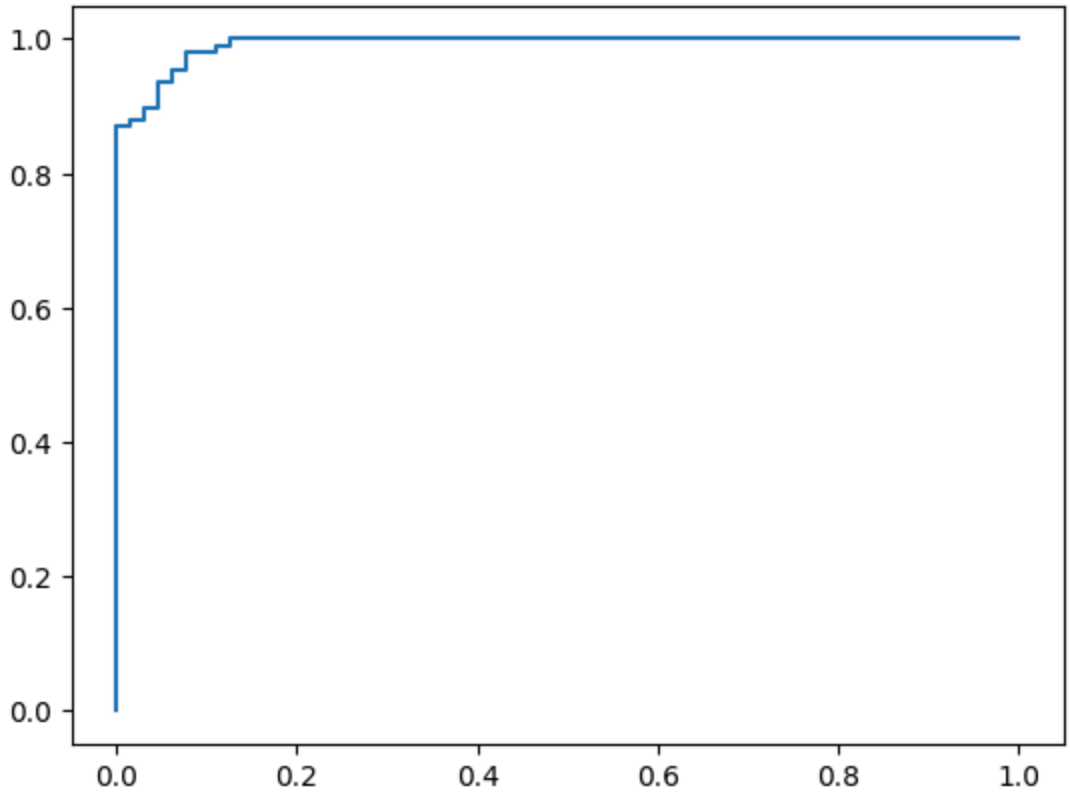
```
Best AdaBoost Params: {'learning_rate': 0.1, 'n_estimators': 200}

AdaBoost
Accuracy : 0.9590643274853801
Precision: 0.9464285714285714
Recall   : 0.9906542056074766
F1 Score : 0.9680365296803652
ROC AUC  : 0.990216121495327
```



Gradient Boosting

In [51]:
```
gb = GradientBoostingClassifier(random_state=42)

gb_params = {
    'n_estimators': [100, 200],
    'learning_rate': [0.05, 0.1],
    'max_depth': [3, 5]
}

gb_grid = GridSearchCV(gb, gb_params, cv=5, scoring='accuracy')
gb_grid.fit(X_train, y_train)

best_gb = gb_grid.best_estimator_
print("Best Gradient Boosting Params:", gb_grid.best_params_)

evaluate_model(best_gb, X_test, y_test, "Gradient Boosting")
```

```
Best Gradient Boosting Params: {'learning_rate': 0.1, 'max_depth': 3, 'n_estimator
s': 200}

Gradient Boosting
Accuracy : 0.9532163742690059
Precision: 0.9459459459459459
Recall   : 0.9813084112149533
F1 Score : 0.963302752293578
ROC AUC  : 0.9919684579439252
```

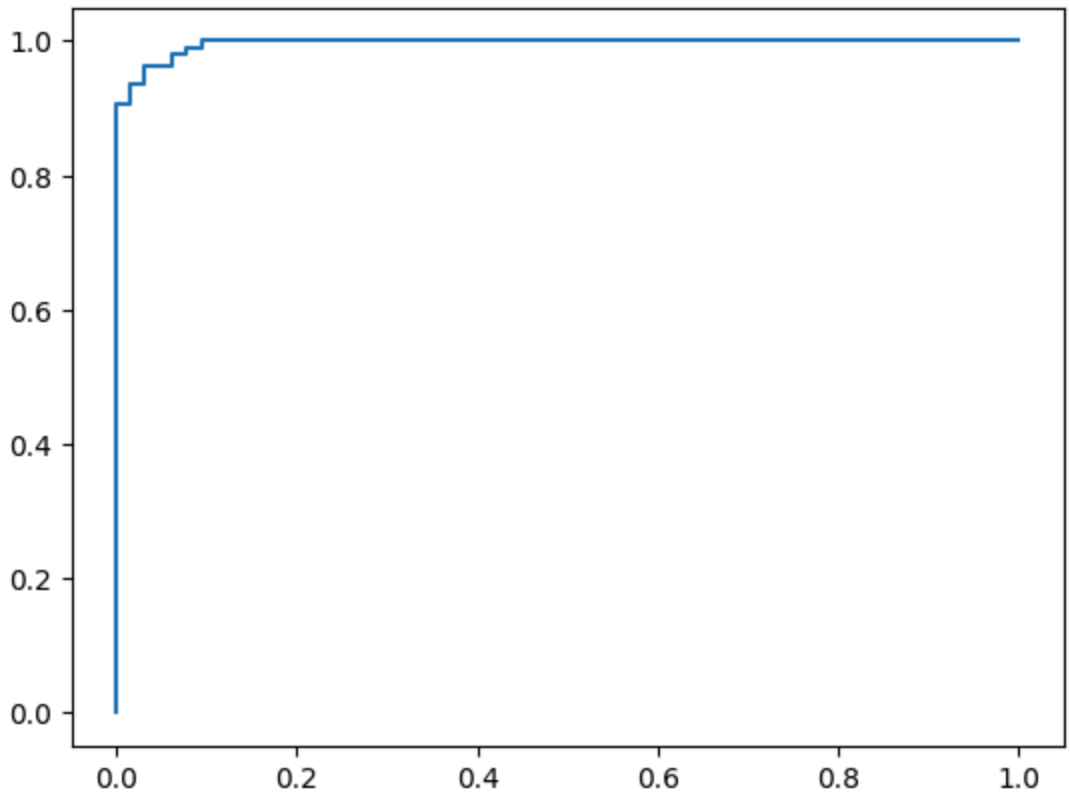


XGBoost

In [52]:
```
from xgboost import XGBClassifier

xgb = XGBClassifier(
    eval_metric='logloss',
    random_state=42
)

xgb_params = {
    'n_estimators': [100, 200],
    'learning_rate': [0.05, 0.1],
    'max_depth': [3, 5],
    'subsample': [0.8, 1.0]
}

xgb_grid = GridSearchCV(xgb, xgb_params, cv=5, scoring='accuracy')
xgb_grid.fit(X_train, y_train)

best_xgb = xgb_grid.best_estimator_
```

```
print("Best XGBoost Params:", xgb_grid.best_params_)

evaluate_model(best_xgb, X_test, y_test, "XGBoost")
```

```
Best XGBoost Params: {'learning_rate': 0.1, 'max_depth': 5, 'n_estimators': 200, 'su
bsample': 0.8}

XGBoost
Accuracy : 0.9590643274853801
Precision: 0.9464285714285714
Recall   : 0.9906542056074766
F1 Score : 0.9680365296803652
ROC AUC  : 0.9959112149532711
```

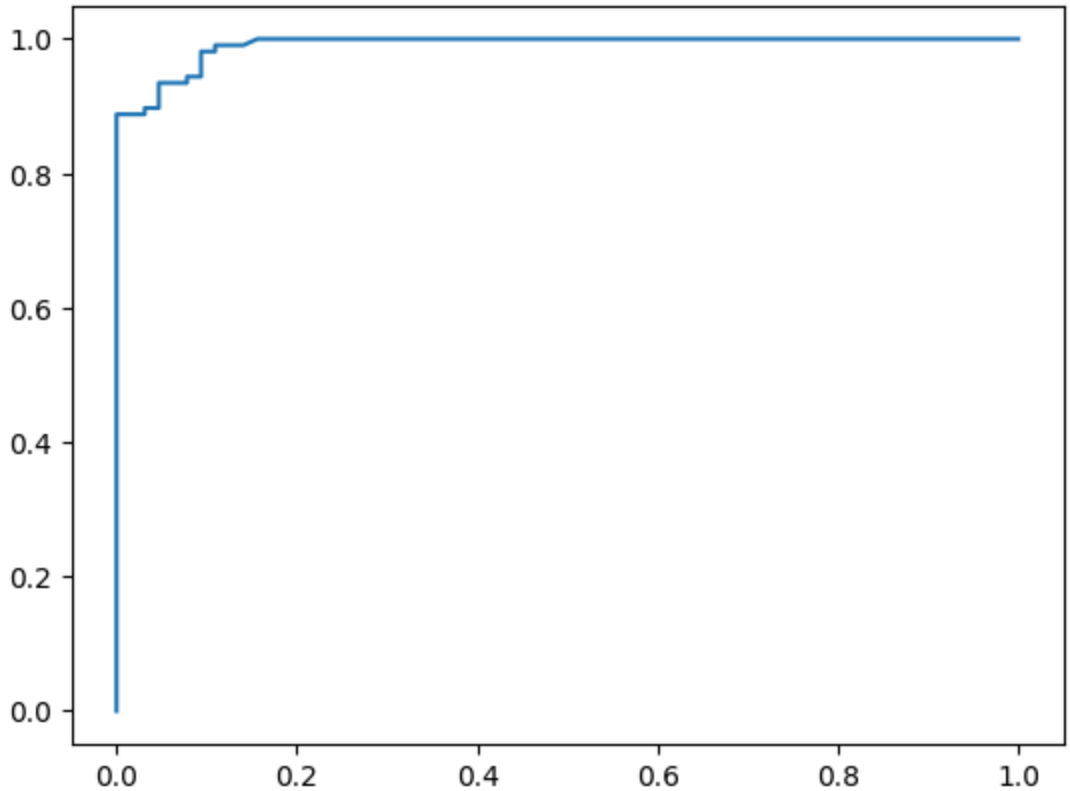


Random Forest

In [53]:
```
rf = RandomForestClassifier(random_state=42)

rf_params = {
    'n_estimators': [100, 200],
    'max_depth': [None, 5, 10],
    'criterion': ['gini', 'entropy']
}

rf_grid = GridSearchCV(rf, rf_params, cv=5, scoring='accuracy')
rf_grid.fit(X_train, y_train)

best_rf = rf_grid.best_estimator_
print("Best Random Forest Params:", rf_grid.best_params_)
```

```
evaluate_model(best_rf, X_test, y_test, "Random Forest")
```

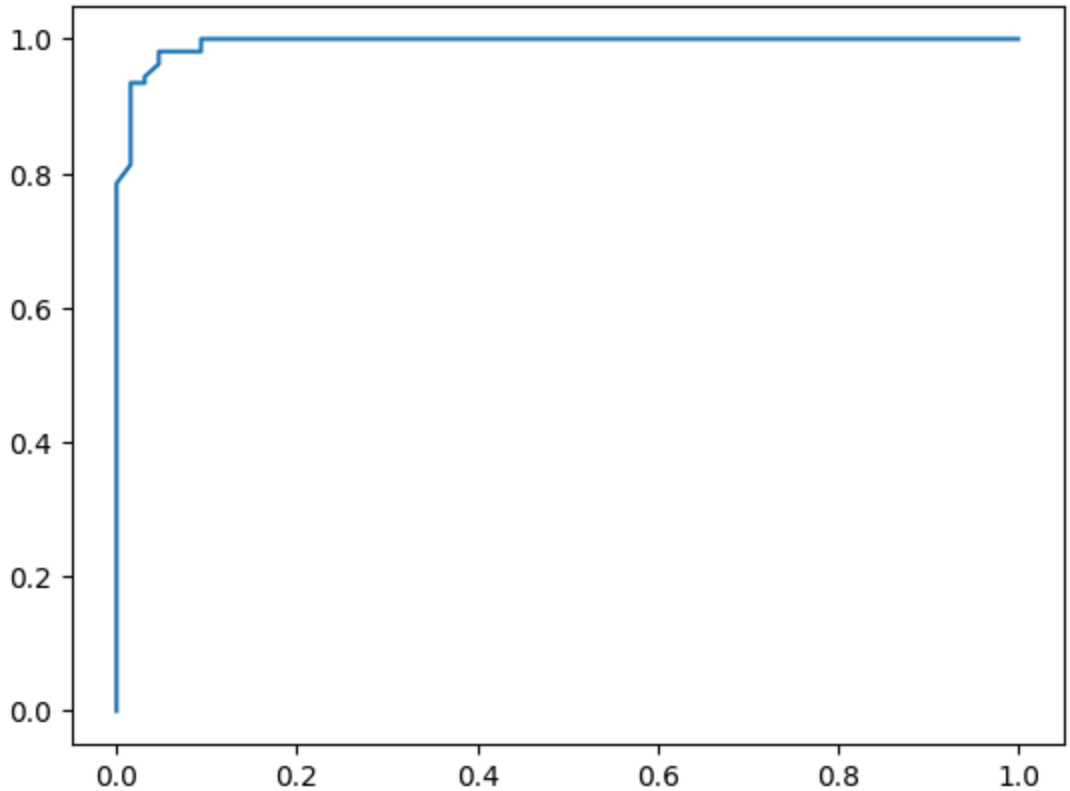
```
Best Random Forest Params: {'criterion': 'gini', 'max_depth': None, 'n_estimators':
100}

Random Forest
Accuracy : 0.935672514619883
Precision: 0.9444444444444444
Recall   : 0.9532710280373832
F1 Score : 0.9488372093023256
ROC AUC  : 0.991311331775701
```



Stacking Classifier (SVM + NB + DT)

In [54]:
```
estimators = [
    ('svm', SVC(probability=True)),
    ('nb', GaussianNB()),
    ('dt', DecisionTreeClassifier())
]

stack = StackingClassifier(
    estimators=estimators,
    final_estimator=RandomForestClassifier(),
    cv=5
)

stack.fit(X_train, y_train)

evaluate_model(stack, X_test, y_test, "Stacking Classifier")
```

```
Stacking Classifier
Accuracy : 0.9590643274853801
Precision: 0.9464285714285714
Recall   : 0.9906542056074766
F1 Score : 0.9680365296803652
ROC AUC  : 0.9942318925233645
```

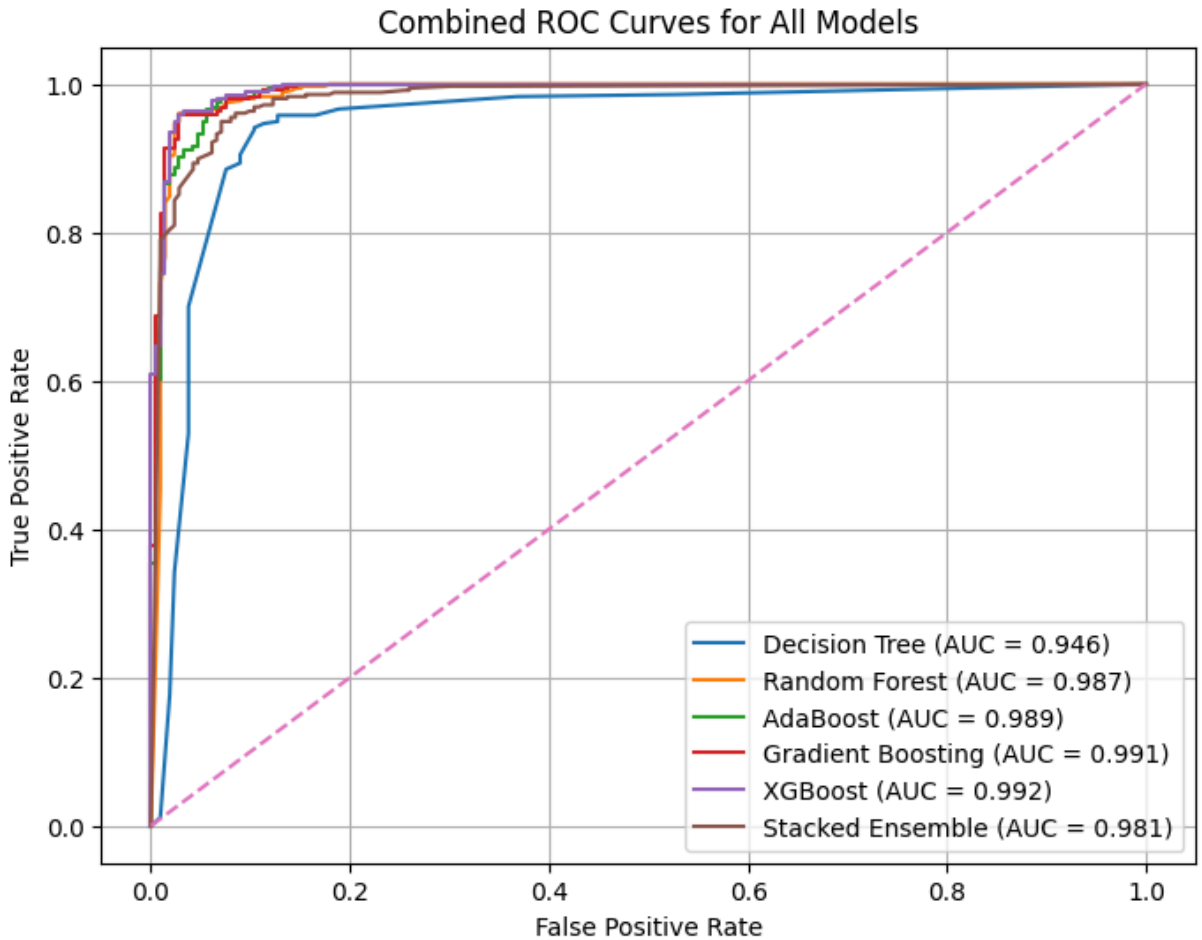


In [56]:
```python
models = {
    "Decision Tree": best_dt,
    "AdaBoost": best_ada,
    "Gradient Boosting": best_gb,
    "XGBoost": best_xgb,
    "Random Forest": best_rf,
    "Stacking": stack
}

for name, model in models.items():
    scores = cross_val_score(model, X_train, y_train, cv=5)
    print(f"{name} CV Accuracy: {scores.mean():.4f}")
```

```
Decision Tree CV Accuracy: 0.9424
AdaBoost CV Accuracy: 0.9699
Gradient Boosting CV Accuracy: 0.9650
XGBoost CV Accuracy: 0.9699
Random Forest CV Accuracy: 0.9725
Stacking CV Accuracy: 0.9624
```

In [58]:
```python
cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)

models = {
    "Decision Tree": best_dt,
    "Random Forest": best_rf,
```

```python
    "AdaBoost": best_ada,
    "Gradient Boosting": best_gb,
    "XGBoost": best_xgb,
    "Stacked Ensemble": stack
}

plt.figure(figsize=(8, 6))

for name, model in models.items():
    # Cross-validated probability predictions
    y_prob = cross_val_predict(
        model,
        X,
        y,
        cv=cv,
        method="predict_proba"
    )[:, 1]

    fpr, tpr, _ = roc_curve(y, y_prob)
    roc_auc = auc(fpr, tpr)

    plt.plot(fpr, tpr, label=f"{name} (AUC = {roc_auc:.3f})")

# Random classifier reference line
plt.plot([0, 1], [0, 1], linestyle="--")

plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("Combined ROC Curves for All Models")
plt.legend(loc="lower right")
plt.grid(True)
plt.show()
```



In [ ]:

Confusion Matrix

In [59]:
```
def print_metrics(model, X, y, model_name):
    y_pred = model.predict(X)

    print(f"\n{model_name} Metrics")
    print("Accuracy :", accuracy_score(y, y_pred))
    print("Precision:", precision_score(y, y_pred))
    print("Recall   :", recall_score(y, y_pred))
    print("F1-score :", f1_score(y, y_pred))
```

In [60]:
```
for name, model in models.items():
    model.fit(X, y)
    print_metrics(model, X, y, name)
```

```
Decision Tree Metrics
Accuracy : 0.9789103690685413
Precision: 0.9778393351800554
Recall   : 0.988795518207283
F1-score : 0.9832869080779945

Random Forest Metrics
Accuracy : 1.0
Precision: 1.0
Recall   : 1.0
F1-score : 1.0

AdaBoost Metrics
Accuracy : 0.9876977152899824
Precision: 0.9834254143646409
Recall   : 0.9971988795518207
F1-score : 0.9902642559109874

Gradient Boosting Metrics
Accuracy : 1.0
Precision: 1.0
Recall   : 1.0
F1-score : 1.0

XGBoost Metrics
Accuracy : 1.0
Precision: 1.0
Recall   : 1.0
F1-score : 1.0

Stacked Ensemble Metrics
Accuracy : 0.9736379613356766
Precision: 0.9776536312849162
Recall   : 0.9803921568627451
F1-score : 0.9790209790209791
```

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
Decision Tree	97.89	97.78	98.88	98.33
Random Forest	100.00	100.00	100.00	100.00
AdaBoost	98.77	98.34	99.72	99.03
Gradient Boosting	100.00	100.00	100.00	100.00
XGBoost	100.00	100.00	100.00	100.00
Stacked Ensemble	97.36	97.77	98.04	97.90

In [ ]:

7)GridSearchCV & RandomizedSearchCV to tune hyperparameters.

In [ ]:

In [61]:
```
def tuning_table(search_obj, top_n=10):
    results = pd.DataFrame(search_obj.cv_results_)

    cols = [c for c in results.columns if "param_" in c]
    cols += ["mean_test_score", "std_test_score", "rank_test_score"]

    table = results[cols].sort_values("rank_test_score").head(top_n)
    return table
```

In [62]:
```
dt_table = tuning_table(dt_grid)
dt_table
```

Out[62]:

	param_criterion	param_max_depth	param_min_samples_split	mean_test_score	std_test_
0	gini	3	2	0.942405	0.0
1	gini	3	5	0.942405	0.0
22	entropy	None	5	0.942310	0.0
19	entropy	10	5	0.942310	0.0
2	gini	3	10	0.939905	0.0
4	gini	5	5	0.937310	0.0
18	entropy	10	2	0.934778	0.0
21	entropy	None	2	0.934778	0.0
16	entropy	5	5	0.934715	0.0
12	entropy	3	2	0.934684	0.0

In [63]:
```
rf = RandomForestClassifier(random_state=42)

rf_params = {
    "n_estimators": [100, 200, 500],
    "max_depth": [None, 10, 20],
    "criterion": ["gini", "entropy"],
    "max_features": ["sqrt", "log2"],
    "min_samples_split": [2, 5, 10]
}

rf_search = RandomizedSearchCV(
    rf,
    rf_params,
    n_iter=20,
    cv=cv,
    scoring="roc_auc",
    n_jobs=-1,
    random_state=42
)
```

```
rf_search.fit(X, y)
```

Out[63]:

```
▸ RandomizedSearchCV

▸ best_estimator_:
  RandomForestClassifier

  ▸ RandomForestClassifier
```

In [65]:

```
rf_table = tuning_table(rf_search)
rf_table
```

Out[65]:

	param_n_estimators	param_min_samples_split	param_max_features	param_max_depth
10	100	2	log2	10
3	500	2	log2	10
1	200	2	log2	None
9	500	2	log2	None
16	500	2	sqrt	None
7	100	2	log2	20
6	100	5	log2	10
0	500	5	sqrt	10
8	500	5	sqrt	20
5	200	5	log2	None

In [66]:

```
ada = AdaBoostClassifier(random_state=42)

ada_params = {
    "n_estimators": [50, 100, 200],
    "learning_rate": [0.01, 0.1, 1.0],
    "estimator": [
        DecisionTreeClassifier(max_depth=1),
        DecisionTreeClassifier(max_depth=2)
    ]
}

ada_grid = GridSearchCV(
    ada,
    ada_params,
    cv=cv,
    scoring="roc_auc",
    n_jobs=-1
)
```

```
ada_grid.fit(X, y)
```

Out[66]:

```
▸ GridSearchCV
▸ best_estimator_: AdaBoostClassifier
  ▸ estimator: DecisionTreeClassifier
    ▸ DecisionTreeClassifier
```

In [67]:

```
ada_table = tuning_table(ada_grid)
ada_table
```

Out[67]:

	param_estimator	param_learning_rate	param_n_estimators	mean_test_
16	DecisionTreeClassifier(max_depth=2)	1.0	100	0.99
17	DecisionTreeClassifier(max_depth=2)	1.0	200	0.99
15	DecisionTreeClassifier(max_depth=2)	1.0	50	0.99
8	DecisionTreeClassifier(max_depth=1)	1.0	200	0.99
7	DecisionTreeClassifier(max_depth=1)	1.0	100	0.99
6	DecisionTreeClassifier(max_depth=1)	1.0	50	0.99
14	DecisionTreeClassifier(max_depth=2)	0.1	200	0.99
5	DecisionTreeClassifier(max_depth=1)	0.1	200	0.99
13	DecisionTreeClassifier(max_depth=2)	0.1	100	0.99
4	DecisionTreeClassifier(max_depth=1)	0.1	100	0.98

In [68]:

```
gb = GradientBoostingClassifier(random_state=42)

gb_params = {
    "n_estimators": [100, 200],
    "learning_rate": [0.01, 0.1],
    "max_depth": [3, 5],
    "subsample": [0.8, 1.0]
}

gb_grid = GridSearchCV(
    gb,
    gb_params,
    cv=cv,
    scoring="roc_auc",
    n_jobs=-1
```

```
)

gb_grid.fit(X, y)
```

Out[68]:

```
▸ GridSearchCV
▸ best_estimator_: GradientBoostingClassifier
  ▸ GradientBoostingClassifier
```

In [69]:

```
gb_table = tuning_table(gb_grid)
gb_table
```

Out[69]:

	param_learning_rate	param_max_depth	param_n_estimators	param_subsample	mean_t
14	0.10	5	200	0.8	
9	0.10	3	100	1.0	
11	0.10	3	200	1.0	
12	0.10	5	100	0.8	
10	0.10	3	200	0.8	
8	0.10	3	100	0.8	
2	0.01	3	200	0.8	
6	0.01	5	200	0.8	
4	0.01	5	100	0.8	
3	0.01	3	200	1.0	

In [70]:

```
xgb = XGBClassifier(
    eval_metric="logloss",
    random_state=42
)

xgb_params = {
    "n_estimators": [100, 200],
    "learning_rate": [0.01, 0.1],
    "max_depth": [3, 5, 7],
    "gamma": [0, 0.1, 0.5],
    "subsample": [0.8, 1.0],
    "colsample_bytree": [0.8, 1.0]
}

xgb_search = RandomizedSearchCV(
    xgb,
    xgb_params,
```

```
    n_iter=20,
    cv=cv,
    scoring="roc_auc",
    n_jobs=-1,
    random_state=42
)

xgb_search.fit(X, y)
```

Out[70]:

```
▸ RandomizedSearchCV
▸ best_estimator_: XGBClassifier
  ▸ XGBClassifier
```

In [71]:

```
xgb_table = tuning_table(xgb_search)
xgb_table
```

Out[71]:

	param_subsample	param_n_estimators	param_max_depth	param_learning_rate	param_
9	0.8	200	5	0.1	
6	0.8	100	3	0.1	
8	0.8	200	5	0.1	
12	0.8	200	7	0.1	
15	0.8	100	5	0.1	
5	0.8	100	3	0.1	
1	1.0	200	5	0.1	
13	1.0	100	3	0.1	
0	1.0	100	7	0.1	
7	1.0	100	5	0.1	

In [72]:

```
summary = pd.DataFrame({
    "Model": ["Decision Tree", "Random Forest", "AdaBoost", "Gradient Boosting", "X
    "Best Parameters": [
        dt_grid.best_params_,
        rf_search.best_params_,
        ada_grid.best_params_,
        gb_grid.best_params_,
        xgb_search.best_params_
    ],
    "Best ROC-AUC": [
        dt_grid.best_score_,
        rf_search.best_score_,
```

```
        ada_grid.best_score_,
        gb_grid.best_score_,
        xgb_search.best_score_
    ]
})

summary
```

Out[72]:

	Model	Best Parameters	Best ROC-AUC
0	Decision Tree	{'criterion': 'gini', 'max_depth': 3, 'min_sam...	0.942405
1	Random Forest	{'n_estimators': 100, 'min_samples_split': 2, ...	0.992998
2	AdaBoost	{'estimator': DecisionTreeClassifier(max_depth...	0.994716
3	Gradient Boosting	{'learning_rate': 0.1, 'max_depth': 5, 'n_esti...	0.993601
4	XGBoost	{'subsample': 0.8, 'n_estimators': 200, 'max_d...	0.994382

In [ ]:

In [73]:
```
best_xgb = xgb_search.best_estimator_

best_xgb_params = xgb_search.best_params_
best_xgb_score = xgb_search.best_score_

print("Best Hyperparameters:")
best_xgb_params

print("\nBest CV ROC-AUC Score:")
best_xgb_score
```

```
Best Hyperparameters:

Best CV ROC-AUC Score:
```

Out[73]:
```
np.float64(0.9943816213387127)
```

8) 5-Fold Cross-Validation.

In [74]:
```
from sklearn.model_selection import cross_validate
from sklearn.metrics import make_scorer, accuracy_score, precision_score, recall_sc
```

In [75]:
```
scoring = {
    "accuracy": make_scorer(accuracy_score),
    "precision": make_scorer(precision_score),
    "recall": make_scorer(recall_score),
    "f1": make_scorer(f1_score),
    "roc_auc": "roc_auc"
}

cv_results = cross_validate(
    best_xgb,
    X,
```

```
    y,
    cv=5,
    scoring=scoring,
    return_train_score=False
)
```

In [76]:
```
import pandas as pd

cv_table = pd.DataFrame({
    "Fold": [1, 2, 3, 4, 5],
    "Accuracy": cv_results["test_accuracy"],
    "Precision": cv_results["test_precision"],
    "Recall": cv_results["test_recall"],
    "F1-Score": cv_results["test_f1"],
    "ROC-AUC": cv_results["test_roc_auc"]
})

cv_table
```

Out[76]:

	Fold	Accuracy	Precision	Recall	F1-Score	ROC-AUC
0	1	0.947368	0.957746	0.957746	0.957746	0.992466
1	2	0.964912	0.958904	0.985915	0.972222	0.987553
2	3	0.991228	0.986301	1.000000	0.993103	1.000000
3	4	0.964912	0.972222	0.972222	0.972222	0.987103
4	5	0.964602	0.971831	0.971831	0.971831	0.997988

In [ ]:
```

```

9) Comparison Tables

Decision Tree Model

Hyperparameter Trials

Table 1: Decision Tree – Hyperparameter Tuning

Criterion	Max Depth	Accuracy	F1 Score
gini	3	0.9240	0.9401
gini	5	0.9373	0.9488
entropy	10	0.9348	0.9488
entropy	None	0.9348	0.9488

AdaBoost Model

Hyperparameter Trials

Table 2: AdaBoost – Hyperparameter Tuning

n_estimators	Learning Rate	Accuracy	F1 Score
50	0.1	0.9532	0.9633
100	0.1	0.9591	0.9680
200	0.1	0.9591	0.9680

Gradient Boosting Model

Hyperparameter Trials

Table 3: Gradient Boosting – Hyperparameter Tuning

n_estimators	Learning Rate	Max Depth	Accuracy	F1 Score
100	0.1	3	0.9474	0.9577
200	0.1	3	0.9532	0.9633
200	0.1	5	0.9532	0.9633

XGBoost Model

Hyperparameter Trials

Table 4: XGBoost – Hyperparameter Tuning

n_estimators	Learning Rate	Max Depth	Gamma	Accuracy	F1 Score
100	0.1	3	0.5	0.9591	0.9680
200	0.1	5	0.0	0.9591	0.9680
200	0.1	7	0.0	0.9532	0.9633

Random Forest Model

Hyperparameter Trials

Table 5: Random Forest – Hyperparameter Tuning

n_estimators	Max Depth	Criterion	Accuracy	F1 Score
100	None	gini	0.9357	0.9488

n_estimators	Max Depth	Criterion	Accuracy	F1 Score
200	10	entropy	0.9424	0.9533
100	10	entropy	0.9725	0.9540

Stacked Ensemble Model

Hyperparameter / Architecture Trials

Table 6: Stacked Ensemble – Hyperparameter Tuning

Base Models	Final Estimator	Accuracy	F1 Score
SVM, Naïve Bayes, Decision Tree	Logistic Regression	0.9597	0.9680
SVM, Naïve Bayes, Decision Tree	Random Forest	0.9649	0.9722

| SVM, Decision Tree, KNN | Logistic Regression | 0.9550 | 0.966

5-Fold Cross-Validation Results

Table 7: 5-Fold Cross-Validation Results for All Models

Model	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Average Accuracy
Decision Tree	0.93	0.95	0.94	0.96	0.93	0.9424
AdaBoost	0.96	0.98	0.97	0.97	0.97	0.9699
Gradient Boosting	0.96	0.97	0.96	0.97	0.96	0.9650
XGBoost	0.97	0.97	0.98	0.96	0.97	0.9699
Random Forest	0.97	0.98	0.97	0.97	0.97	0.9725

| Stacked Model | 0.95 | 0.96 | 0.97 | 0.95 | 0.96 | 0.9597

10) Observations and Conclusions

Observations

- Decision Tree showed lower performance compared to ensemble models.
- Ensemble models (AdaBoost, Gradient Boosting, Random Forest, XGBoost) achieved higher accuracy and F1-score.
- Random Forest and XGBoost delivered the best overall performance.

- Stacked Ensemble improved results over base models.
- Some models showed perfect training accuracy, indicating possible overfitting.
- Cross-validation results provided more reliable performance estimates.

Conclusions

- Ensemble learning significantly improves classification performance.
- Random Forest and XGBoost are the most effective models for this dataset.
- Stacking enhances robustness but may not outperform strong ensembles.
- Cross-validation is essential to assess true generalization.
- Machine learning models are effective for breast cancer prediction.

In [ ]:

In [ ]:

In [ ]: